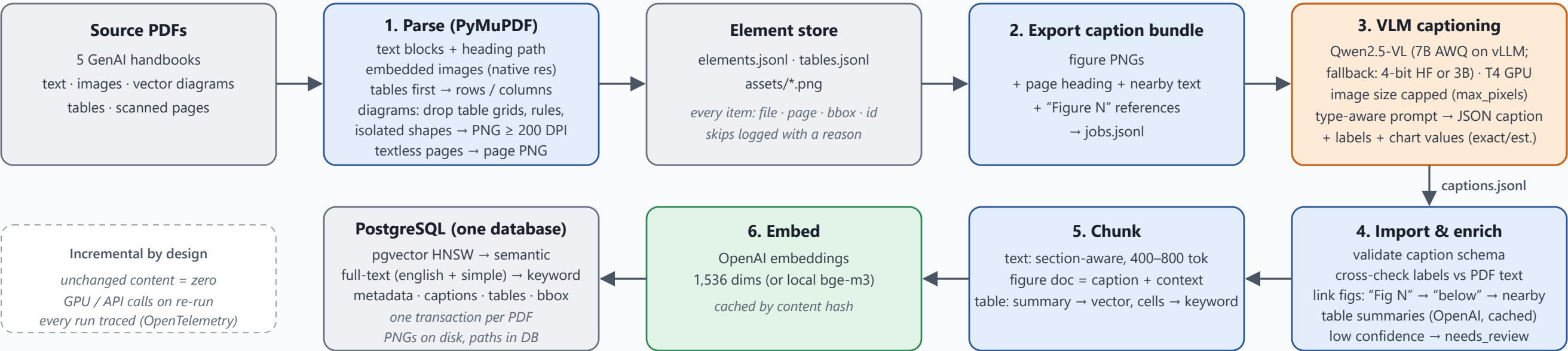


Agentic Multimodal RAG: Application Flow

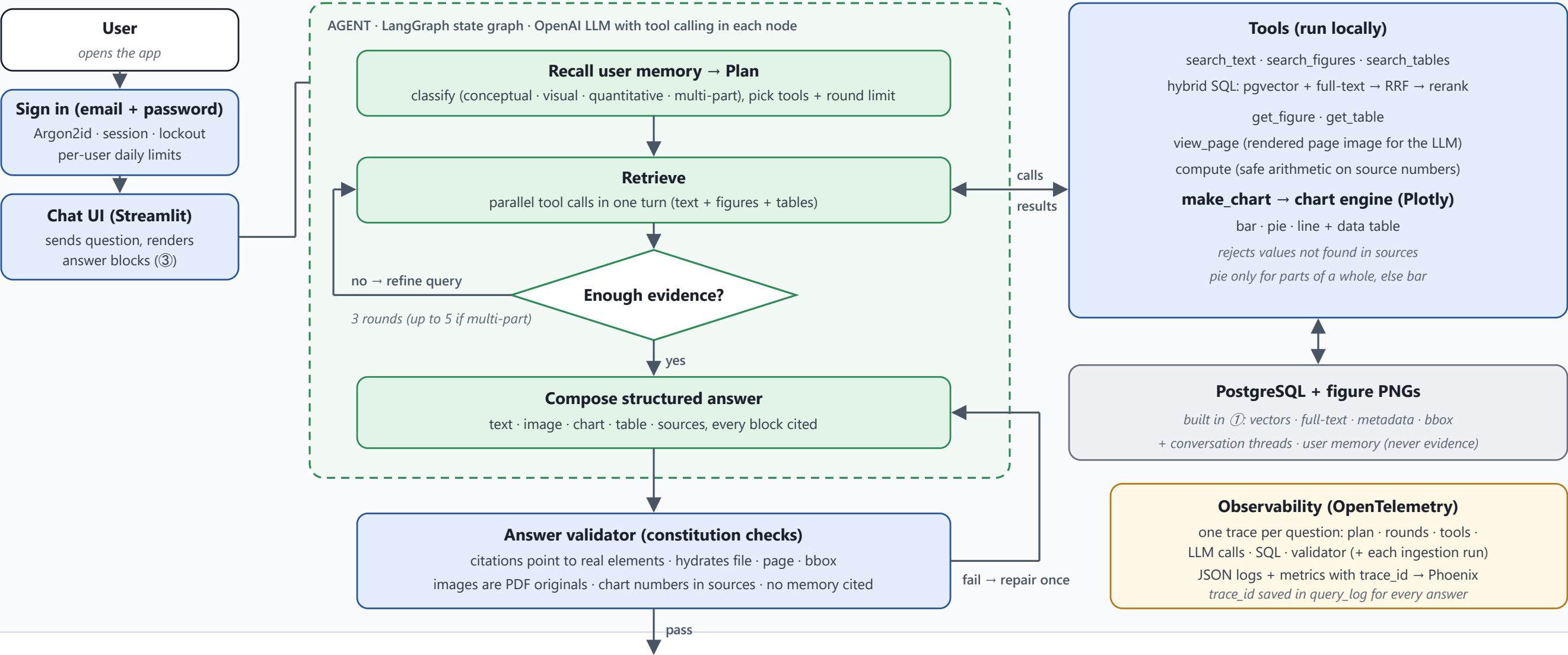
GPU used only for VLM captioning at ingestion · OpenAI API for reasoning at query time · PostgreSQL (pgvector + full-text) stores everything · rest runs on the laptop CPU

Laptop CPU Temporary GPU (Kaggle / Colab) OpenAI API Data store Answer output

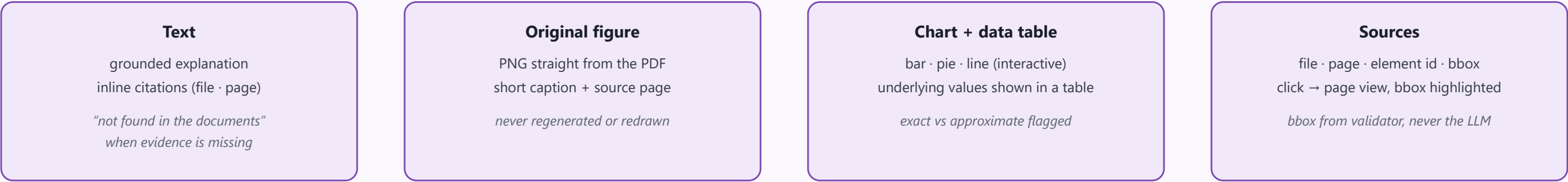
1 INGESTION: offline, run once per PDF



2 QUERY: online, every question (no GPU)



3 MULTIMODAL ANSWER: rendered in the chat UI



4 EVALUATION: offline, before any change is kept (RAGAS + custom checks)

